Unit 9 Advanced Programming			
Unit aims	This unit enables learners to focus on the popular Object-Oriented paradigm and associated SOLID principles as exemplified in programming languages such as Java, C++, C# and Python. It includes a strong promotion of the DevSecOps practice; the notion of security being "shifted to the left" to avoid costly baked-in bugs, encouraging a shared responsibility through the software's lifecycle and its quality assurance processes. Furthermore, it encourages learners to consider modern approaches (and tooling) used when products are designed, written, and then deployed at scale in potentially costly (and high-profile) software engineering projects.		
Unit level	5		
Unit code	Y/651/0158		
GLH	60		
Credit value	15		
Unit grading structure	Pass, Merit and Distinction		
Assessment guidance	Tutors will need to help learners set up suitable development, (potentially) staging and production environments, and make available suitable open-source automation servers and utilities. This may be achieved "on-prem", i.e. without use of cloud or hybrid services. In order to achieve this unit, learners must produce work which demonstrates achievement of the learning outcomes at the standards provided by the assessment criteria. For a Merit, all pass and all merit criteria must be met. For a Distinction, all pass, all merit and all distinction criteria must be met.		
Learning outcomes The learner will demonstrate that they:	**Assessment criteria** The learner can:		
	Pass	Merit	Distinction
1. Understand the importance of implementing SOLID principles on the OOP paradigm	1.1 Explain the aim of the 5 key SOLID principles and the many ways code can be evaluated to ensure compliance with design patterns 1.2 Define the four pillars of Object-Oriented Programming with a coded example for each	1M1 Design and implement a solution for a complex problem which uses all four pillars of OOP	1D1 Design and Implement a solution for a complex problem which requires two or more SOLID principles applied

2. Understand the importance of DevSecOps as a set of practices, cultural approaches, and tools in software development	2.1 Explain the importance of implementing Shift Left as a DevSecOps best practice 2.2 Explain the problems DevSecOps aims to solve during the development lifecycle	2M1 Design suitable SCA, SAST and DAST stages using appropriate tools and configuration	2D1 Implement a DevSecOps CI/CD pipeline using an open-source automation server
3. Can design, build and test a large dataset application using SOLID principles, clean coding techniques, and automated testing	3.1 Demonstrate designing and building a large dataset to a client's requirement 3.2 Describe examples of SOLID principles and clean coding techniques applied as part of the solution 3.3 Build an automated test to a set specification 3.4 Explain the benefits of automated testing over manual testing	3M1 Demonstrate the utilisation of automated testing tools	

Indicative Content

1. Understand the importance of implementing SOLID principles on the OOP paradigm

- Object Orientated Programming (OOP)
  - OOP paradigm characteristics (class (encapsulation), object (instantiation), properties, methods, inheritance, constructors, destructors, virtual classes, overriding, access modifiers etc.)
  - Object-oriented class relationships, e.g.
    - Inheritance
    - Realisation/Implementation
    - Composition
    - Aggregation
    - Association
    - Dependencies
  - Use of standard notation, visual modelling to communicate OO ideas, e.g. UML
  - Clean coding techniques, including use of design patterns, SOLID etc
- SOLID design principles:
  - Understanding and applying the component parts of SOLID design principles is important to make software that is understandable, extensible, and maintainable
  - To include the following five common principles:
    - Single responsibility principle (SRP)

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        - Open-closed principle
        - Liskov substitution principle
        - Interface segregation principle
        - Dependency inversion principle

2. Understand the importance of DevSecOps as a set of practices, cultural approaches, and tools in software development

- DevSecOps
    - Multidisciplinary nature, e.g. software engineers (developers), Security, Operations
    - Secure development lifecycle surrounding 8 discrete DevOps stages
    - Dev
        - Plan
        - Code
        - Build
        - Test
    - Ops
        - Release
        - Deploy
        - Operate
        - Monitor
    - Defined security patterns for the services and applications built
    - Ending with automated security for automated operations
- DevSecOps - best implementation practices
    - Cultural shift and impact
    - Threat Modelling; risks to systems and data
    - "Shift Left" concept, i.e. implementing security as early as possible
    - Continuous Integration and Continuous Delivery (CI/CD); key components
    - CI/CD open-source automation server, e.g. Jenkins, Gitlab etc.
    - Types of security tests:
        - Source Composition Analysis (SCA)
        - Static Application Security Testing (SAST)
        - Dynamic Application Security Testing (DAST)
    - Obfuscation Techniques; protecting your code using uglification, minification etc.
    - Microservices Architecture; advantages (development, testing, deployment)
    - Cloud-native Technologies; Containers (Docker, Kubernetes – K8s), Microservices etc.
    - Role-based Access Control (RBAC) security model; types and rationale
    - Log Activity, Monitor, Regular Audit, and Review

3. Design, build, and test a large dataset application using SOLID principles, clean coding techniques and automated testing

- Datasets
    - Developing and handling of large datasets
    - Data structures
    - Data operations
    - Data implementation
        - Collect the raw data
        - Identify feature and label sources
        - Select a sampling strategy
        - Split the data
- Testing
    - Types of automatic testing
    - Tool automation parameters

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- Follow common testing frameworks and methodologies
  - Tools
  - Self-built testing tools

Suggested Resources

Mellor, Alan, 2021, Java OOP Done Right: Creative object oriented code you can be proud of with modern Java, Self-publishing

Kalb and Irv, 2022, Object-Oriented Python: Master OOP by Building Games and GUIs, No Starch Press

Lott, Steven F., & Phillips, Dusty., 2021, Python Object-Oriented Programming: Build robust and maintainable object-oriented Python applications and libraries, 4th Edition, Packt Publishing

Mack, Sean D., 2023, The DevSecOps Playbook: Delivery Continuous Security at Speed, Wiley

Wilson, Glenn, 2020, DevSecOps: A leader's guide to producing secure software without compromising flow, feedback and continuous improvement, Self-publishing

McLean Hall, Gary, 2017, Adaptive Code: Agile coding and design patterns and SOLID principles (Developer Best Practices) Microsoft Press

Websites

www.skillsyouneed.com
www.coursera.org
www.geeksforgeeks.org/introduction-of-object-oriented-programming/
www.educative.io/blog/object-oriented-programming
www.tutorialspoint.com/human_computer_interface/object_oriented_programming.htm
www.ibm.com/topics/devsecops
www.redhat.com/en/topics/devops/what-is-devsecops
https://aws.amazon.com/what-is/devsecops/
www.geeksforgeeks.org/solid-principle-in-programming-understand-with-real-life-examples/
www.digitalocean.com/community/conceptual-articles/s-o-l-i-d-the-first-five-principles-of-object-oriented-design
www.freecodecamp.org/news/solid-principles-explained-in-plain-english/

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Supporting information	
Opportunities for Synoptic Teaching and Learning	
Learners and tutors will have the opportunity to link the learning from this unit with the content of other units.	
Learning Outcome	**Teaching and learning links to other unit LO/AC**
LO1 Understand the importance of implementing SOLID principles on the OOP paradigm	This unit from the **Software** pathway together with **Unit 6** (**LO1, LO2** and **LO3**) in the **Data** pathway and **Unit 13** (**LO1, LO2** and **LO3**) in the **Cyber** pathway are all interlinked because learners explore programming concepts and how scripting languages are used in contexts such as security. **U8 LO1** and **U8 LO3** are particularly relevant because they involve code and algorithms. It is likely that learners will use Python as their language of choice in the **Software** pathway and equally it is likely that Python will be used in the **Data** units. This is because Python is one of the most popular languages because it is so versatile and it is often used when working with AI, statistics and it is also used in Penetration Testing and Ethical Hacking: **U13 LO1**, **U13 LO2**, **U13 LO3**. This does not mean that C++, C and languages such as Java do not feature, it just means that for many contexts, Python may be the language of choice.
LO2 Understand the importance DevSecOps as a set of practices, cultural approaches, and tools in software development	
LO3 Design, build and test a large dataset application using SOLID principles, clean coding techniques, and automated testing	
Opportunities for Synoptic Assessment	
Unit 17 Advanced Project provides an opportunity for learners to demonstrate an understanding of the concepts of this unit when learners use the skills learned in both the core and their chosen pathway to resolve a business problem or show how a business opportunity could be pursued using appropriate tools and technologies. The project should be a suitable match to their study pathway and should make use of the knowledge and skills gained when studying the Diploma and Extended Diploma units.	

Opportunities for Skills Development within this unit	
Employability Skills	**Study Skills**
Problem-solving – Understanding the application of SOLID principles on the OOP paradigm (LO1). Recognising the importance of DevSecOps and the reasons why software development should embrace security at the point of design and implementation (LO2). Correctly using development tools and practices in the development of secure applications (LO1, LO2, LO3). **Communication** - Written communication, e.g. appropriate formats, style and tone, spelling, punctuation &	**Time management** – Identifying and sticking with your objective, setting yourself realistic goals (LO3). **Using sources of information** – Accessing information, e.g. choosing current, sufficient, relevant and reliable sources (LO1, LO2, LO3). **Reading Skills** – Identifying different sources of information, reading with a purpose (LO1, LO2, LO3). **Thinking Skills** – Using critical thinking skills, e.g. reading all information, identifying and

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grammar (SPAG) (LO1, LO2, LO3). Oral communication, e.g. presenting and choosing appropriate presentation formats (LO3). **Working independently** - Using own initiative, self-organisation, maintaining motivation/focus, managing tasks (LO1, LO2, LO3). **Digital Skills** – Using appropriate digital devices, handling and judging the reliability of information, problem-solving (eg online research, presenting data) (LO1, LO2, LO3).	interpreting others' positions, arguments and conclusions, weighing up opposing arguments, drawing conclusions (LO1, LO2, LO3). **Writing Skills** - Understanding the purpose of writing; understanding writing styles and forms; ensuring that written output is suitable for the intended audience (LO1, LO2, LO3).